

# Curriculum vitae



**DR. SOURAV BHADWAL**  
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**Submitted Ph.D. (Geography)**  
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## Objective:

To achieve success in all sphere of profession by holding a position of responsibility in my career accepting every challenge and to serve the humanity through my abilities.

## Educational Qualifications:

| Degree                                    | Year | Institute (Board)                                | Details                                                                                                                    |
|-------------------------------------------|------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Ph.D. in (Geography)                      | 2025 | Central University of Haryana                    | <b>Topic:</b> <i>Assessment and Evaluation of Green Space for Sustainable Urban Development: A Case Study of NCT Delhi</i> |
| Post Graduate Diploma in (Geoinformatics) | 2025 | Indira Gandhi National Open University (IGNOU)   | Pursuing                                                                                                                   |
| Master of Science (Geography)             | 2019 | Central University of Haryana                    | First Division                                                                                                             |
| Bachelor of Science                       | 2017 | Govt. Degree College, Kathua<br>Jammu University | First Division                                                                                                             |

**Area of research domain** includes the application of remote sensing and GIS in urban sustainability, ecosystem services, and environmental monitoring, with a focus on urban green spaces, green infrastructure, carbon sequestration, urban heat islands, land use and land cover dynamics, and climate change adaptation.

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## Academic Achievements:

- **Qualified NTA-NET** in Geography
- **Received Best Paper Presentation Award** in the Technical Session “*Earth Observation (EO) and Future Earth*” at **International Geographical Union (IGU) Thematic Conference** held on 24<sup>th</sup> -25<sup>th</sup> Nov, 2022. Sponsored by Indian Council of Social Science Research (ICSSR).
- Attended a workshop on Remote Sensing by **NASA<sub>s</sub> (ARSET)**
- Seven Off-Campus Certificate Courses organized by **Indian Institute of Remote Sensing** sponsored by **(ISRO)**
- Total 30 + certificates of **(National, state & Inter university levels)** Co-curricular activities.

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## Technical Skills

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- **Geospatial and Remote Sensing Software:**  
ArcGIS, ERDAS IMAGINE, IBM SPSS
  - **Programming & Data Analytics:**  
Python (data analysis, visualization, machine learning),  
R Language (statistical analysis, modelling, data visualization)  
Google Earth Engine
- **Other Skills:**  
Data handling and cleaning, scientific writing, research documentation, spatial data visualization, quantitative and qualitative analysis

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## Working Experience

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- Worked as a **Research Assistant (2023-2025)** under **ICSSR Major Research Project** entitled, “**Water management in south-western Haryana**”. File No. ICSSR/RPD/MJ/2023-24/G/49, Department of Geography, School of Basic Sciences in Central University of Haryana.
- Worked as a **Project Assistant** under the sponsored research project entitled, “**Remote Sensing and GIS Based Modelling of Land Degradation for Agricultural Sustainability in Sultanpur District, Uttar Pradesh**”, sanctioned by Research and Development Plan 2000, Higher Education Department, Uttar Pradesh Government (Letter No. 46/2021/603/Sattar 4-20-4 (56), in the Department of Geography, Rana Pratap P.G. College, Sultanpur, U.P
- Worked as a **Project Assistant** under National Institute of Educational Planning and Administration (NIEPA) Project entitled, “**Skill Formation and Employability: A Study of Youth in India**”. Project Code No. P0040, Department of Teacher Education, School of Education in Central University of Haryana.

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## Paper in Scopus/Web of science Indexed Journals

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1. **Sourav Bhadwal**, Syed Irtiza Majid, & Manish Kumar (2026). “*Quantification and Mapping of Carbon Storage and Sequestration Ecosystem Service in Urban Parks of NCT Delhi: Geo-Environmental Determinants and Sustainability Implications*”. **Geography and Sustainability (Impact Factor: 8.0) JCR - Q1 (under review)**
2. Manish Kumar, **Sourav Bhadwal**, & Netrananda Sahu (2025). “Role of Urban Green Space in Regulating Land Surface Temperature in NCT Delhi using Explainable Artificial Intelligence”. **Urban Climate 65 (2026): 102736**. <https://doi.org/10.1016/j.uclim.2025.102736> **(Impact Factor: 6.9). JCR – Q1.**
3. Manish Kumar, Dinesh Kumar Tripathi, **Sourav Bhadwal**, and Syed Irtiza Majid (2025). “Behavioral assessment and modelling of land degradation using random forest regression models and SHAP-based Game theory”. **European Journal of Soil Science, 76(6)**. <https://doi.org/10.1111/ejss.70224> **(Impact**

**Factor: 3.8). JCR – Q1**

4. Manish Kumar, Alka Dash, Syed Irtiza Majid, Akash Tiwari, **Sourav Bhadwal**, and Netrananda Sahu (2025). “Delineation of groundwater potential zones of India using GIS based Fuzzy-AHP techniques”. **Environmental and Sustainability Indicators**, 28, 100932. <https://doi.org/10.1016/j.indic.2025.100932> **(Impact Factor: 5.6). JCR – Q1**
5. Syed Irtiza Majid, Manish Kumar, & **Sourav Bhadwal** (2025). “Application of explainable artificial intelligence to decode water-induced soil erosion in Lidder watershed of the Greater Himalayas”. **Stochastic Environmental Research and Risk Assessment**, 39, 3903–3936. <https://doi.org/10.1007/s00477-025-03042-9> **(Impact Factor: 3.6). JCR – Q1**
6. Pragya, Manish Kumar, Akash Tiwari, Syed Irtiza Majid, **Sourav Bhadwal**, Netrananda Sahu, Naresh Kumar Verma, Dinesh Kumar Tripathi, and Ram Avtar (2023). "Integrated spatial analysis of forest fire susceptibility in the indian Western Himalayas (IWH) using remote sensing and GIS-based fuzzy AHP approach." **Remote Sensing**, 15, no. 19: 4701. <https://doi.org/10.3390/rs15194701> **(Impact Factor: 4.1). JCR - Q1**
7. Kumar, Ravi, Manish Kumar, Akash Tiwari, Syed Irtiza Majid, **Sourav Bhadwal**, Netrananda Sahu, and Ram Avtar (2023). "Assessment and Mapping of Riverine Flood Susceptibility (RFS) in India through Coupled Multicriteria Decision Making Models and Geospatial Techniques." **Water**, 15, no. 22: 3918. <https://doi.org/10.3390/w15223918> **(Impact Factor: 3.0). JCR - Q2**
8. Manish Kumar, Dinesh Kumar Tripathi, Syed Irtiza Majid, & **Sourav Bhadwal** (2025). “*A novel GIS-based approach for modelling soil erosion susceptibility using integrated empirical and machine learning models.*” **Environmental, Development and Sustainability** **(Impact Factor: 4.2) JCR - Q1 (under review)**
9. Akash Tiwari, Manish Kumar, Syed Irtiza Majid, **Sourav Bhadwal**, Naresh Kumar Verma, Dinesh Kumar Tripathi and Subhash Anand (2024). “Unravelling the Role of Socio-Physical Drivers for Potential Built-up Site Selection in the Kumaun Himalayas Using GIS-Based Fuzzy-AHP and Machine Learning”. **Journal of Settlements and Spatial Planning**, vol. 15, no. 1 (2024) 23-38. <https://doi.org/10.24193/JSSP.2024.1.03> **(Impact Factor: 0.7). JCR – Q3**

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## **Book Chapters in Springer**

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1. **Sourav Bhadwal**, Naresh Kumar Verma, Sanjit Kumar, Manish Kumar, Pankaj Kumar (2022). Morphometric Analysis and Geohydrological Inference of Bhilangna Drainage Basin, Uttarakhand (India) Using Remote Sensing and GIS Techniques, Water Resources Management and Sustainability pp 403-422, **Springer International Publishing**, ISBN 978-981-16-6572-1, [https://doi.org/10.1007/978-981-16-6573-8\\_19](https://doi.org/10.1007/978-981-16-6573-8_19).

2. Kumar, Sanjit, **Bhadwal, Sourav, et al.** "Spatio-Temporal Changes in Metropolitan Cities of India: A Comparative Study of Delhi and Mumbai." Remote Sensing and Geographic Information Systems for Policy Decision Support. **Springer, Singapore**, 2022. 265-282. [https://doi.org/10.1007/978-981-16-7731-1\\_12](https://doi.org/10.1007/978-981-16-7731-1_12)

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## Content writing in M.Sc. Geo-informatics programme in Uttarakhand Open University

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- Written three units (**Unit 4 - Data Manipulation, Unit 6 – Raster Data Formats, Unit 12 – Map Manipulation**) of M.A., M.Sc 2<sup>nd</sup> Semester for the course named “**Advance GIS**” (M.G.I.S-06) of **Geo-informatics programme** run by Uttarakhand Open University, Haldwani.

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## Paper/ Poster Presented in Conferences (Abroad)

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1. 3rd International Workshop on Geographic Modeling and Simulation: Modeling and Simulation for a Sustainable World, held from October 25–27, 2024, entitled ***Geospatial Modelling and Mapping of Groundwater Potential Zones in the Western Himalayan Region of India using GIS-based Fuzzy Analytical Hierarchy Process***. Organized by **Nanjing Normal University Nanjing, China**.
2. Annual Conference “**Recent trends in Climate Sciences, adaptation and mitigation**” organized by **Italian Society for Climate Sciences (SISC), Italy**. Author Topic entitled “**Mapping frequency and Magnitude of cloud burst in the Himalayas & its Simulation**”
3. Annual Conference “Recent trends in Climate Sciences, adaptation and mitigation” organized by **Italian Society for Climate Sciences (SISC), Italy**. Co-Author Topic entitled “**The Response of Himalayan Glaciers to Intrinsic Climate Variability: Little ice Age onwards**”

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## Paper/Poster Presented in Conferences (India)

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1. International Conference on “Geospatial Innovations for Biodiversity Conservation, Climate Resilience, and Sustainable Development Across Multi-Domain Ecosystems,” held on April 8–9, 2025, entitled ***Application of Machine and Deep Learning-based Game Theory for Evaluating the Impact of Green Infrastructure on Air Quality in NCT Delhi, India***. Organized by Kalindi College, University of Delhi, in collaboration with the Centre for Himalayan Studies, University of Delhi.
2. International Geographical Union Thematic Conference Sponsored by ICSSR, “Sustainability, Future Earth and Humanities: Opportunities and Challenges” on 24-25 Nov 2022, entitled ***Spatio-temporal evolution of urban green space and its impact on the urban thermal environment based on remote sensing data: A case study of NCT Delhi*** organized by Department of Geography, Central University of Haryana

3. International conference on “Challenge to Disaster Risk reduction and Resilient Habitat” organized by **Centre for Disaster Management Studies Shaheed Bhagat Singh college, New Delhi**. Author topic entitled *“Spatio-temporal dynamics of Urban Green Space A case study of Delhi”*
4. International conference on “Food, Water and Nutrition Security for Health and wellbeing” organized by **Department of Geography, Saheed Bhagat Singh college, New Delhi**. Author topic entitled *“Evaluation and Analysis of Fluoride content in drinking water and prevalence of fluorosis by using geo-spatial techniques: A case study of Mahendergarh, Haryana”*
5. Presented a paper in National Conference Sponsored by SERB, “Natural Resources Sustainability & Geospatial Technology” on 21-22<sup>nd</sup> April 2023, entitled ***Urban Green Space: Challenges and Prospects*** organized by Department of Geography, GDC Memorial College, Bahal, Bhiwani, Haryana.
6. Presented a paper in National seminar Sponsored by ICSSR, “Economy @ 75: Azadi Ka Amrit Mahotsav” on 16 Nov 2022, entitled ***Urbanization and Diminishing Urban green space: A study through Geospatial Approach*** Organized by Department of Economics, Central University of Haryana.
7. Presented a paper in Two Day National Seminar Sponsored by ICSSR, “75 Years of Economics Development: Women Entrepreneurship for Sustainability” on 10-11 Oct 2022, entitled ***Gender Disparities in Exposure to Green Space: An Empirical Study of South Delhi***, Organized by Central University of Haryana.
8. National Seminar on “Smart City, Rural Development and Sustainable clean Environment organized by RPS College, Mahendragarh. (Oral), Topic entitled **“Mapping Seasonal & Temporal Dynamic of Urban Heat Island (UHI)”**
9. National Seminar on “BHARAT @ 75: Social, Economic, Political and Cultural Dimensions organized by Central University of Haryana, Mahendragarh. (Oral), Topic entitled **“Spatio-temporal variation of COVID-19: District level analysis of Rajasthan using GIS & Remote Sensing.”**
10. Annual conference on “Heading towards Zero: Sustainable Development in Economy, Environment and Society” organized by **International Geographical Union (IGU), India**. Author Topic entitled *“Morphological analysis of Bhilangna River, Tehri District, Uttarakhand using Remote Sensing.*
11. Annual conference on “Heading towards Zero: Sustainable Development in Economy, Environment and Society” organized by **International Geographical Union (IGU), India**. Author Topic entitled *“Mapping Seasonal and Temporal dynamics of Urban Heat Island (UHI) of Faridabad City”*

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### **Participation in Webinars:**

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1. Participated in One Day International Webinar on “**Future of Geoinformatics technology**” organized by Department of Geography, CBLU, Bhiwani and Technical Partner Netra Institute of Geoinformatics Management & Technology Foundation Dawarka, New Delhi.
2. Participated in the Webinar Organized by the Department of Geography, Adamas University on the theme of “**Application of GIS-RS Based Techniques and Research Horizon in Geographical Studies.**”
3. Participated in one day Webinar on “**Empathetic Conversational Artificial Intelligence (AI)**” organized by University of Delhi.
4. Participated in the online webinar programme on “**Landslide risk reduction and resilience**” organized by **National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India** in collaboration with **Geological Survey of India, Ministry of Mines, Govt. of India.**
5. Participated in the online webinar programme on “**Climate change and extreme Weather Events**” organized by **National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India** in collaboration with **Indian Meteorological Department, Ministry of Earth Sciences, Govt. of India.**
6. Participated in the online webinar programme on “**Earthquake Monitoring and Management**” organized by **National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India** in collaboration with **National center of Seismology, Ministry of Earth Sciences, Govt. of India.**
7. Participated in the online webinar programme on “**GeSpatial Application in Disaster Management**” organized by **National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India** in collaboration with **IIT, Tirupati.**
8. Attended an International Webinar on “**Challenges of Identifying Geographic Trends in COVID-19 Cases & Effect of COVID-19 Lockdown on Air-quality in India**”. Organized by University of Baroda.

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### **E-Trainings / Programmes / Workshops:**

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1. Participated in **21 Days DST Workshop “Summer School on Geospatial Sciences and technologies (Level 1)”** from 23 May to 12 June 2022, Organized by Department of Geography, Chaudhary Bansi Lal University, Bhiwani.
2. Participated in **7 days national workshop on” WebGIS Technology and Application in Resource Management using FOSS4G”** from 6-12 April 2022, sponsored by SERB, organized by Department of Geography, Burdwan University, West Bengal.
3. Participated in **7 days national workshop on “Application of Statistical tools in Data Analysis (ASTDA)”** from 21-25 March 2022, organized by Department of Statistics, Central University of

Haryana.

4. Completed **21 days** online “**GIS Training Programme using QGIS**”, Conducted by the Department of Geography, University of Karnataka, jointly with State Institute of Urban Development, Karnataka.
5. Completed **15- Days NASA’s** Applied Remote Sensing Training Programme on the topic “**Using Earth Observation to Monitor Water Budgets for River Basin Management**”
6. Completed **13- Days** Course on “**Basic Principles of Remote Sensing Technology**” conducted by Indian Institute of Remote Sensing (IIRS), ISRO Dehradun.
7. Completed **7- Days online Summer School** on “**Advances in Signal Processing and Machine learning**” organized by University of Delhi.
8. Participated in a **4-Day online Faculty Development Programme** on “**PYTHON PROGRAMMING**” organized by Department of computer science& Engineering, K L Deemed to be University Hyderabad in association with IIT Bombay spoken Tutorials, funded by NMEICT, MHRD, Govt. of India.
9. Participated in **2 Days Extensive Workshop** on “**INTRODUCTION TO PYTHON PROGRAMMING**” organized by Progressio Alas.
10. Completed **Three- day** e- training on “**Basics of Image Interpretation**” conducted PGRS Division of Geological Survey of India Training Institute, Hyderabad.
11. Participated in the **3-Days** online Training Programme on “**Climate Change, Landslides and Safe Hill Area Development**” organized by National Institute of Disaster Management, Ministry of Home Affairs, Govt. of India in Collaboration with Dr. Raghunandan Singh Tolia Uttarakhand Academy of administration, Nanital, Uttarakhand.
12. Completed **13 hours and 30 Minutes** online course on “**Overview of Geoprocessing using Python**” conducted by IIRS, Dehradun

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### **Strengths:**

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- Punctual
- Confident
- Disciplined
- Hard working
- Flexible in nature